Chapter 6 — Electronic Measurements

RRB JE CBT-2 · Electronics & Allied Engineering (CEN 05/2025, Annexure-VII C) · Syllabus Unit 6

> **Syllabus scope:** Measuring systems; Basic principles of measurement; Range Extension methods; Cathode Ray Oscilloscope, LCD, LED panel; Transducers.

📊 Weightage & Trend Box

Item	Detail
Approximate weight	~4% → **2 questions out of 50** in the actual CBT-2 paper of 22-04-2025
Observed 2025 Q1	Why is the oscilloscope probe cable **shielded (coaxial)**? → To reduce pickup of **external electrical interference/noise (EMI)** on the small signal being carried to the CRO.
Observed 2025 Q2	On what does the length (delay) of an **electrical delay line** in a CRO depend? → On the **signal propagation speed**, which is fixed by the **L and C parameters** of the lumped LC line (delay per section = √(LC)).
Trend	Questions are **conceptual, one-liner, CRO-centric** so far — but the unit officially covers instruments, range extension, bridges, displays and transducers, all of which are standard JE-level question banks. Expect shunt/multiplier numericals, LVDT/thermocouple facts, and Lissajous figures in future papers.
Strategy	Master (i) CRO block diagram + probes, (ii) Rsh and Rs formulas, (iii) transducer one-liners (LVDT, gauge factor, Seebeck, PT100, NTC), (iv) the four AC bridges and what each measures.

6.1 Measuring Systems — The Generalized Measurement System

A **measurement** is the process of comparing an unknown quantity (the *measurand*) with a predefined, accepted **standard**. Every practical instrument — from a simple ammeter to a DSO — can be decomposed into the same chain of **functional elements**:

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Measurand → [Primary Sensing Element] → [Variable Conversion Element]
          → [Signal Conditioning Element] → [Data Transmission]
          → [Data Presentation Element] → Observer
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Functional element	Job	Example
Primary sensing element	First element to receive energy from the measurand; produces an output that depends on the measurand	Bourdon tube (pressure), thermocouple junction, strain gauge grid
Variable conversion element	Converts the sensed variable into a more convenient form (often electrical) without changing the information	Displacement → inductance change (LVDT core), resistance → voltage (bridge)
Signal conditioning element	Amplifies, filters, attenuates, modulates, integrates or otherwise manipulates the signal to suit the next stage	Instrumentation amplifier, low-pass filter, A/D converter
Data transmission element	Carries the conditioned signal to the display/recording point	Shielded cable, 4–20 mA current loop, telemetry link
Data presentation element	Presents the result to the observer in readable/usable form	Pointer & scale, seven-segment LED, LCD, chart recorder, CRO screen

Exam nuggets

- The **primary sensing element** is also called the **detector**; it is the *transducer* in most electronic systems.
- The order **sensing → conversion → conditioning → presentation** is a favourite one-mark sequencing question.
- Data may also pass through **data processing** (computation, linearization) before presentation in modern digital instruments.

6.2 Basic Principles of Measurement

6.2.1 Accuracy, Precision, Resolution, Sensitivity

Term	Meaning	Key point
Accuracy	Closeness of the measured value to the **true value**	Expressed as % of full-scale deflection (FSD) or % of reading
Precision	Closeness of **repeated measurements to each other** (repeatability/reproducibility)	An instrument can be precise but inaccurate (consistent offset)
Resolution	**Smallest change** in input that the instrument can detect/display	For a 3½-digit DVM on 2 V range: resolution = 1 mV
Sensitivity	Ratio of **change in output** to **change in input**: S = Δoutput/Δinput	Slope of the static calibration curve; for a voltmeter, sensitivity = 1/I_fsd in **Ω/V**
Threshold	Minimum input below which no output change is seen (starting from zero)	—
Dead zone / dead band	Largest change of input to which the instrument does not respond	Caused by friction, backlash, hysteresis
Drift	Gradual shift of output with time at constant input	Zero drift, span drift
Hysteresis	Different output for the same input depending on whether input is increasing or decreasing	Magnetic/mechanical causes
Linearity	Closeness of the calibration curve to a straight line	Non-linearity quoted as % of FSD

> ☆ **Accuracy vs precision picture:** darts clustered tightly away from bullseye = *precise, not accurate*; darts scattered around the bullseye = *accurate on average, not precise*.

6.2.2 Errors in Measurement

Classification (very frequent MCQ):

1. **Gross errors** — human mistakes: misreading the scale, parallax, wrong recording, wrong instrument choice. *Remedy:* care, multiple readings by different observers. They **cannot be treated mathematically**.
2. **Systematic errors** — consistent, repeatable, of definite sign:

- **Instrumental errors** — inherent shortcomings (weak spring, friction, worn parts), misuse (not zeroing), and **loading effect** (instrument alters the quantity it measures — e.g., a low-sensitivity voltmeter across a high resistance).
- **Environmental errors** — temperature, humidity, pressure, external magnetic/electrostatic fields. *Remedy:* shielding, temperature compensation, controlled conditions.
- **Observational errors** — parallax (remedy: mirror-backed scales), estimation habits.

1. **Random (residual) errors** — remain after gross and systematic errors are removed; caused by many small unpredictable effects. They follow the **Gaussian (normal) distribution** and are treated statistically (mean, standard deviation, probable error = 0.6745 σ).

Error definitions

- **Absolute error:** $\delta A = A_m - A_t$ (measured − true)
- **Relative / percentage error:** $\varepsilon = \delta A / A_t \times 100\ \%$
- **Static correction:** $\delta C = A_t - A_m = -\delta A$
- **Limiting (guarantee) error:** manufacturer guarantees error within ± x % of **full scale**. Hence, the *percentage error of a reading is worst at low readings* — always use an instrument in the **upper third of its scale**.

> **Worked Example 6.1 (limiting error).**

> A 0–150 V voltmeter has a guaranteed accuracy of ±1 % of full scale. It reads 75 V. Find the limiting error in the reading.

> Absolute limiting error = 1 % × 150 = ±1.5 V.

> % limiting error at 75 V = 1.5/75 × 100 = **±2 %**.

> (If it read 15 V, the error would be 1.5/15 = ±10 % — this is why we avoid the low end of the scale.)

> **Worked Example 6.2 (absolute & % error).**

> True value of a resistor is 100.0 Ω; a bridge measures 100.4 Ω.

> Absolute error δA = 100.4 − 100.0 = **+0.4 Ω**; % error = 0.4/100 × 100 = **+0.4 %**; correction = **−0.4 Ω**.

Combination of errors (quantities in a product/quotient): if $X = A \cdot B / C$, then

% error in X ≈ (%err A) + (%err B) + (%err C) (worst-case limiting errors add).

6.2.3 Standards and Calibration

Standard	Description
International standards	Defined by international agreement (SI); maintained at BIPM, Sèvres; not available for ordinary use
Primary standards	Maintained by national labs (NPL New Delhi, NIST); used to calibrate secondary standards
Secondary standards	Reference standards in industrial labs; periodically sent to national labs for certification
Working standards	Day-to-day lab/shop-floor calibration tools

Calibration = comparing the instrument against a standard of known accuracy and adjusting/recording the deviations. Calibration removes/quantifies **systematic** errors; it cannot remove random errors.

6.2.4 Static vs Dynamic Characteristics

- **Static characteristics** — apply when the measurand is constant or slowly varying: accuracy, precision, sensitivity, resolution, drift, linearity, hysteresis, dead zone.
- **Dynamic characteristics** — apply to rapidly varying inputs:
- **Speed of response** — rapidity with which the instrument responds.
- **Fidelity** — faithfulness of output waveform to input changes.
- **Lag** — delay in response (retardation lag, time-delay lag).
- **Dynamic error** — difference between true time-varying value and indicated value (with static error removed).
- Standard test inputs: **step, ramp, sinusoidal**; first-order instruments characterized by **time constant τ**, second-order by **natural frequency ω_n and damping ratio ξ**.

6.3 Classification of Instruments

Basis	Types
Nature of standard	**Absolute instruments** — give the value in terms of physical constants and dimensions (tangent galvanometer, Rayleigh current balance); no calibration needed. **Secondary instruments** — must be calibrated against absolute/standard instruments; all practical meters (ammeter, voltmeter, wattmeter).
Function	**Indicating** (pointer & scale — ammeter, voltmeter), **Recording** (continuous trace vs time — strip-chart recorder, X-Y recorder), **Integrating** (totalize over time — **energy meter**, ampere-hour meter).
Signal handling	**Analog** — output varies continuously (PMMC meter); **Digital** — output in discrete steps/digits (DMM, DVM, frequency counter).
Power source	**Deflection type** (deflection ∝ measurand) vs **null type** (balance is restored; e.g., potentiometer, bridge — more accurate because at balance no current is drawn).

6.4 Operating Principles — The Three Torques

Every analog indicating instrument needs three torques:

1. Deflecting torque (T_d)

Produced by the effect being exploited (magnetic, electrodynamic, electrostatic, thermal, induction, hall). It moves the pointer up-scale.

2. Controlling torque (T_c)

Opposes T_d and brings the pointer to rest where **T_c = T_d**; also returns the pointer to zero when the measurand is removed.

Method	Torque law	Notes
Spring control	T_c = K·θ (∝ deflection)	Two phosphor-bronze spiral springs wound in opposite directions (also serve as current leads in PMMC); instrument can be used in **any position**; scale can be uniform if T_d ∝ I
Gravity control	T_c = W·l·sin θ	Small adjustable weight on the spindle; **must be used vertical/levelled**; cheaper; gives **cramped (non-uniform) scale** at low end since T_c ∝ sin θ

3. Damping torque

Acts only while the pointer **moves**; brings it to rest quickly without oscillation. The instrument should be **just under-damped / critically damped** ("dead-beat").

Method	Mechanism	Used in
Air-friction damping	Light piston in an air chamber, or vane in a sector box	Moving-iron, hot-wire instruments
Fluid-friction damping	Vane submerged in oil	Rare; used where instrument is upright; adds buoyancy support
Eddy-current damping	Aluminium former/disc moving in a permanent-magnet field; induced eddy currents oppose motion (Lenz's law)	**PMMC** (aluminium former), **induction energy meter** (brake magnet on disc). **Most efficient** method; *cannot* be used where the operating field would be disturbed (e.g., moving-iron)
Electromagnetic damping	Circuit closed through low resistance so induced current damps motion	Galvanometers

> ☆ If damping is too strong → **over-damped**, sluggish; too weak → **under-damped**, oscillatory. "Dead-beat" ≈ critically damped.

6.5 PMMC (Permanent Magnet Moving Coil) Instruments

Construction: a light rectangular coil of fine wire wound on an **aluminium former**, pivoted in the uniform radial field of a permanent magnet (with soft-iron core), spring-controlled, eddy-current damped (the former itself provides damping).

Torque equation:

T_d = N·B·I·l·b = N·B·A·I = **G·I** where G = NBA (galvanometer constant)

At balance: $G \cdot I = K \cdot \theta \Longrightarrow \boldsymbol{\theta = (G/K) \cdot I \Longrightarrow \theta \propto I}$ → **uniform (linear) scale**.

Property	PMMC
Works on	**DC only** (with AC, torque reverses each half-cycle → mean torque = 0; pointer just vibrates)
Scale	Uniform
Damping	Eddy-current (Al former)
Control	Spring (springs also lead current in/out)
Sensitivity	Very high (20 kΩ/V typical); low power consumption (25–200 μW)
Errors	Ageing of magnet & springs, temperature (coil Cu resistance +0.4 %/°C — compensated with swamping resistance of manganin), friction
Applications	DC ammeters & voltmeters (with shunt/multiplier), galvanometer, ohmmeter, and — with a rectifier — AC measurement; basis of most analog multimeters

Why DC only? The direction of $T_d = NBAI$ depends on current direction. On AC (50 Hz) the inertia of the movement cannot follow the reversals; average deflecting torque is zero.

6.6 Moving-Iron (MI) Instruments

Two types:

1. **Attraction type** — an eccentrically pivoted soft-iron vane is attracted **into** the field of a fixed coil carrying the current.
2. **Repulsion type** — two soft-iron vanes (one fixed, one movable) inside the coil are **similarly magnetized** by the coil current and repel each other. (Radial-vane and coaxial-vane variants.)

Torque: $T_d = \tfrac{1}{2} I^2 \left(dL/d\theta\right) \Longrightarrow \boldsymbol{\theta \propto I^2}$ (approximately)

Property	Moving iron
Works on	**Both AC and DC** (deflection $\propto I^2$, independent of current direction); on AC it reads **RMS**
Scale	**Non-uniform** — cramped at the lower end (square-law); made more uniform by shaping vanes
Control	Spring or gravity
Damping	**Air friction** (eddy-current damping would need a magnet that distorts the operating field)
Errors	Hysteresis (reads higher on descending DC), stray magnetic fields, frequency error (coil inductance) on AC, temperature
Merits	Cheap, robust, AC/DC, withstands overloads
Uses	Switchboard AC ammeters/voltmeters (most common industrial AC meter)

6.7 Electrodynamometer (Dynamometer) Instruments & Wattmeter

Replace the permanent magnet of PMMC with **fixed current coils**; a movable **pressure (potential) coil** rotates in their field.

- Torque: $T_d \propto i_1 \cdot i_2 \cdot (dM/d\theta)$ where M = mutual inductance between coils.
- As ammeter/voltmeter: both coils carry the same current → $\theta \propto I^2$ → **square-law scale**, reads RMS, works on AC and DC — used as **transfer instruments** (calibrated on DC, used on AC).
- **As a wattmeter (chief use):**
- **Fixed coils = current coils (CC)** in series with load (carry load current, few turns, thick wire).
- **Moving coil = pressure coil (PC)** across the load through a high series resistance (current ∝ voltage, in phase with V).
- Average torque $\propto V \cdot I \cdot \cos\varphi$ = **true power**; **scale is uniform** ($\theta \propto P$).
- Air-friction damping; spring control.
- **Errors:** pressure-coil inductance (reads high at lagging PF), connection error (whether PC is on supply side or load side — compensated wattmeter uses a compensating winding), eddy currents, stray fields.
- **Low power factor wattmeter (LPF):** special design (low PC resistance, compensating coil) for measuring power where cos φ is small (e.g., no-load test of transformer).

6.8 Single-Phase Induction Energy Meter (Integrating Instrument)

Principle: two alternating fluxes — from a **shunt (pressure) magnet** (flux ∝ V, made to lag V by ~90° using a copper shading band) and a **series (current) magnet** (flux ∝ I) — induce eddy currents in a rotating **aluminium disc**. Interaction of each flux with the eddy current of the other produces a driving torque $\propto V \cdot I \cdot \cos\varphi$ = power.

- **Braking:** a permanent **brake magnet** near the disc edge produces braking torque ∝ speed. At steady state: speed **N ∝ power**, so total revolutions ∝ **energy** (kWh). Registered by a geared counting mechanism.
- **Meter constant** = revolutions per kWh.

Errors & adjustments

Error	Cause	Adjustment
Phase (lag) error	Pressure flux not exactly 90° behind V	Lag adjustment: shading band/loop on shunt magnet
Creep	Disc rotates slowly with **only voltage applied, no load** (due to supply overvoltage/vibration/pressure-coil asymmetry)	**Two diametrically opposite holes drilled in the disc** — disc rests when a hole comes under the shunt pole
Speed error	Wrong braking	Move brake magnet radially (in = faster, out = slower)
Friction	Bearing/counter friction, worst at light load	Friction-compensating shading loop gives small constant torque
Overload/temperature/frequency errors	—	Design compensation

> **Worked Example 6.3 (energy meter).**
>
> Meter constant = 1200 rev/kWh. On a 230 V, 10 A, UPF load the disc makes 46 revolutions in 60 s. Is the meter accurate?
>
> Energy actually consumed = 230 × 10 × 1 × (60/3600) = 38.33 Wh = 0.03833 kWh.
>
> Revolutions expected = 1200 × 0.03833 = 46.0. **Meter is reading correctly (0 % error).**

6.9 Rectifier and Thermal Instruments

Rectifier instruments: a PMMC movement fed through a bridge (or half-wave) rectifier measures AC. The PMMC responds to the **average** of the rectified wave but the scale is marked in **RMS assuming a sine wave** (form factor 1.11). Hence, on non-sinusoidal waveforms rectifier meters show **waveform error**. High sensitivity and wide frequency range (up to ~kHz–MHz) — used in analog multimeters for AC ranges.

Thermal instruments (one paragraph): Hot-wire and **thermocouple** instruments use the heating effect I^2R. In the thermocouple type, the current heats a heater wire whose temperature is sensed by a thermocouple driving a PMMC meter. Deflection $\propto I^2$ → they read **true RMS regardless of waveform**, work on AC and DC, and are useful at **radio frequencies** (up to tens of MHz) because there is no coil inductance error. Drawbacks: fragile heater, sluggish, poor overload capacity, cramped square-law scale.

Electrostatic instruments (bonus): force between charged plates, $\theta \propto V^2$; used only as **voltmeters** (especially high voltage); draw negligible current; AC (RMS) and DC.

6.10 Range Extension Methods (High-Yield ★★★)

6.10.1 Ammeter Shunt

A PMMC movement of resistance R_m and full-scale current I_m can measure a larger current I by bypassing the excess through a low resistance **shunt R_sh in parallel**.

Derivation. The shunt and meter are in parallel, so they share the same voltage:

I_sh·R_sh = I_m·R_m, and I_sh = I − I_m.

Define **multiplying power m = I / I_m**. Then

(I − I_m) R_sh = I_m R_m ⟹ R_sh = I_m R_m / (I − I_m)

R_sh = R_m / (m − 1) and **m = 1 + R_m/R_sh**

- Shunts are made of **manganin** (DC) / constantan — very low temperature coefficient, and are 4-terminal (separate current and potential terminals) to avoid contact-resistance error.
- A **swamping resistance** (manganin, ~3×R_m) in series with the coil reduces temperature error.
- **Ayrton (universal) shunt** gives multiple ranges without ever leaving the meter unshunted.

> **Worked Example 6.4 (shunt design).**
>
> A meter has R_m = 50 Ω, I_m = 1 mA. Design a shunt for 1 A full scale.
>
> m = 1 A / 1 mA = 1000. R_sh = 50/(1000 − 1) = **0.05005 Ω ≈ 50.05 mΩ**.

> **Worked Example 6.5.**
>
> The same 1 mA, 50 Ω movement must read 100 mA. m = 100 ⟹ R_sh = 50/99 = **0.505 Ω**. Voltage across the parallel combination at FSD = 1 mA × 50 Ω = 50 mV (all shunted meters are really 50 mV millivoltmeters!).

6.10.2 Voltmeter Multiplier

To measure voltage V with a movement of FSD voltage V_m = I_m R_m, connect a high **series (multiplier) resistance R_s**:

V = I_m (R_m + R_s), with **m = V/V_m**:

$R_s = R_m (m - 1)$ $\Leftrightarrow$ $m = 1 + R_s/R_m$

> **Worked Example 6.6 (multiplier design).**

> Movement: I_m = 1 mA, R_m = 100 Ω (so V_m = 0.1 V). Required range 10 V.

> $m = 10/0.1 = 100 \Longrightarrow R_s = 100 \times (100 - 1) =$ **9900 Ω = 9.9 kΩ**.

6.10.3 Voltmeter Sensitivity (Ω/V) and Loading Effect

Sensitivity S = 1/I_{fsd} (Ω per volt). Total voltmeter resistance on a range = S × (range).

E.g., a 20 kΩ/V meter on the 10 V range presents 200 kΩ.

A voltmeter of finite resistance **loads** the circuit: it lowers the very voltage it is measuring. Loading error is serious in **high-resistance circuits** with **low-sensitivity meters**. DMMs (10 MΩ input) load far less than analog VOMs.

> **Worked Example 6.7 (loading effect).**

> Two 100 kΩ resistors in series across 100 V; true voltage across the lower one = 50 V. A 1000 Ω/V voltmeter on its 50 V range (R_v = 50 kΩ) is connected across the lower resistor.

> Effective lower resistance = 100k ∥ 50k = 33.33 kΩ.

> Reading = 100 × 33.33/(100 + 33.33) = **25 V**.

> % error = (25 − 50)/50 × 100 = **−50 %** (!)

> With a 20 kΩ/V meter (R_v = 1 MΩ): 100k ∥ 1M = 90.9 kΩ; reading = 100 × 90.9/190.9 = 47.6 V; error = −4.8 %. **Higher Ω/V ⇒ less loading.**

6.10.4 Instrument Transformers — CT and PT

For heavy AC currents and high AC voltages, shunts/multipliers become impractical (loss, insulation, safety). **Instrument transformers** step down the quantity and also **isolate** the meter from the high-voltage line.

	Current Transformer (CT)	Potential Transformer (PT/VT)
Connection	**Primary in series** with the line (often 1 turn/bar primary)	**Primary across** the line
Steps	Current down (e.g., 100/5 A, 1000/5 A)	Voltage down (e.g., 11 kV/110 V)
Secondary standard	**5 A (or 1 A)**	**110 V**
Ratio	**Transformation ratio R = I_p/I_s**; nominal ratio = rated I_p/rated I_s; turns ratio n = N_s/N_p	R = V_p/V_s
Errors	**Ratio error** = (nominal ratio − actual ratio)/actual ratio ×100 %; **phase-angle error** (secondary current not exactly 180° from primary) — both caused by magnetizing & core-loss components of excitation current	Ratio and phase-angle errors (usually smaller)
Burden	The load (VA) on the secondary — meters/relays + leads. Quoted in **VA at rated secondary current**. Exceeding rated burden increases errors.	Same concept

> ☆ **Why must a CT secondary NEVER be open-circuited while the primary is energized?**

> The primary current is fixed by the load (CT primary is in series with the line), *not* by the CT. Normally the secondary MMF opposes the primary MMF, keeping core flux small. If the secondary is opened, the **opposing MMF vanishes**, the entire primary MMF magnetizes the core → flux rises enormously → (i) **dangerously high voltage** (kV) induced across the open secondary — lethal to personnel and insulation; (ii) **core saturates and overheats** (huge iron loss); (iii) the core may be left permanently magnetized, ruining accuracy. Therefore CT secondaries are **short-circuited (via a shorting switch) before removing the burden**.

Range extension of wattmeter & energy meter: connect the wattmeter current coil via a **CT** and pressure coil via a **PT**;

Actual power = wattmeter reading × CT ratio × PT ratio (with correction factors for CT/PT phase errors in precision work). Energy meters on HV feeders are similarly operated through CT/PT, and the meter constant is scaled by the same product.

6.11 Measurement of Resistance

Resistances are classed as **low (< 1 Ω)**, **medium (1 Ω – 100 kΩ)**, **high (> 100 kΩ)**.

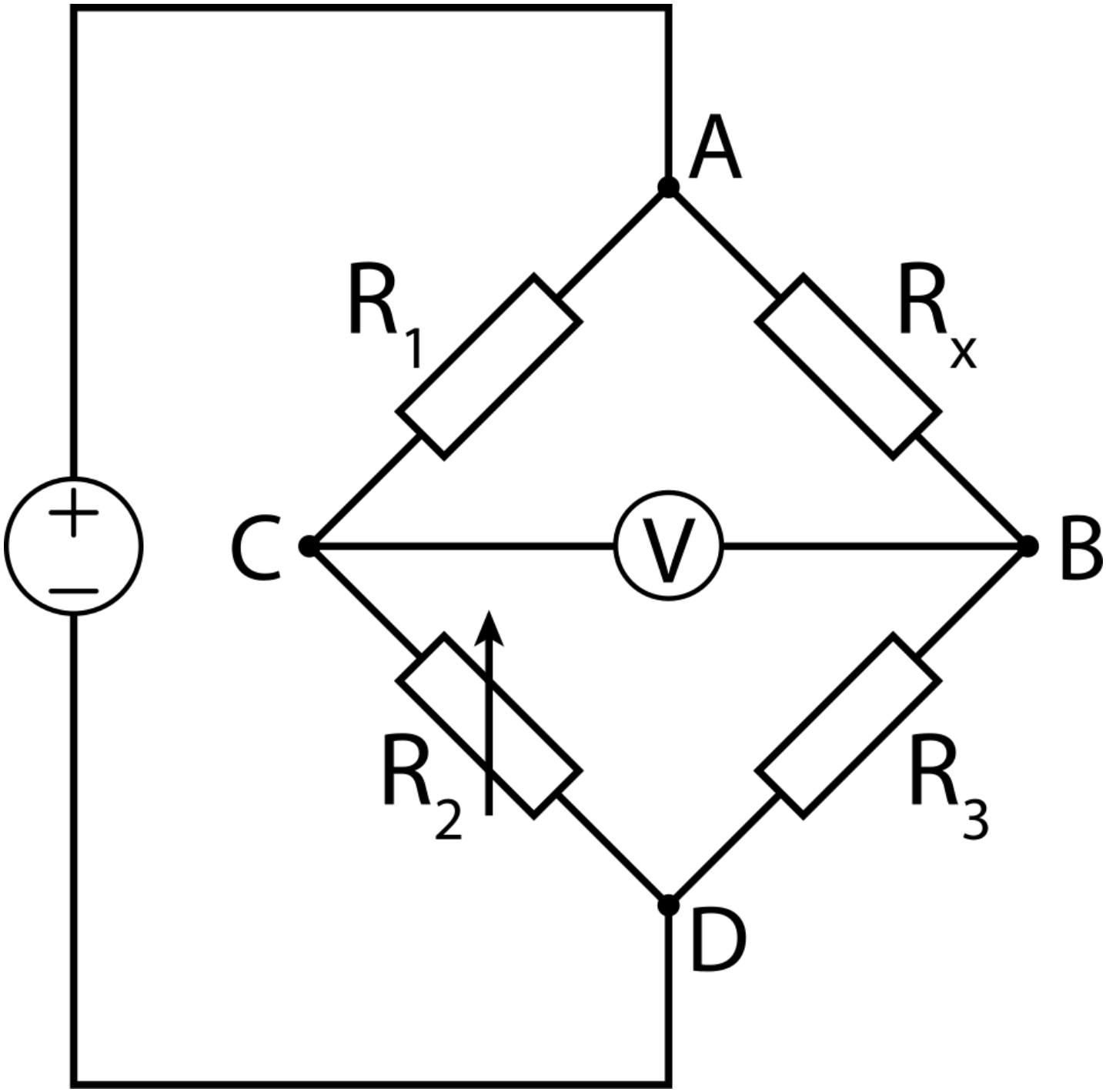
6.11.1 Voltmeter–Ammeter Method (medium R)

R ≈ V/I. Two connections:

- Ammeter near the unknown (voltmeter across both A and R): measured R = R + R_a → good for **high R**.
- Voltmeter directly across R (ammeter reads I_R + I_v): measured R = R ∥ R_v → good for **low R**.

Simple but accuracy limited (~1 %) by meter errors and connection error.

6.11.2 Wheatstone Bridge (medium R) ☆



Wheatstone bridge for resistance measurement.

Four arms P, Q (ratio arms), S (standard variable), R (unknown); galvanometer across the detector diagonal; battery across the other.

Balance condition (no galvanometer current):

Potential at both galvanometer terminals equal ⟹ **P/Q = R/S ⟹ R = (P/Q)·S**

- Null method → high accuracy (0.1 % or better); independent of supply voltage.
- Sensitivity greatest when all four arms are of the same order.
- **Limitations:** for very low R, lead & contact resistances swamp the unknown; for very high R, insulation leakage and low sensitivity spoil it.

6.11.3 Kelvin Double Bridge (low R, 0.1 μΩ – 1 Ω) ☆

A modified Wheatstone with a **second set of ratio arms (p, q)** that eliminates the effect of the **link (yoke) resistance** and connecting-lead resistance. With p/q = P/Q, the balance gives simply

R = (P/Q)·S

and the lead/contact resistances cancel. Unknown and standard are **four-terminal** resistances. Used for shunts, armature/contact resistance, cable strips.

6.11.4 High Resistance / Insulation — Megger ☆

- High R measured by **loss-of-charge method**, **megohm bridge**, or the field instrument: the **megohmmeter (Megger)**.
- **Megger** = a hand-cranked (or battery/electronic) **DC generator (250 V/500 V/1 kV/2.5 kV)** + a **ratiometer (ohmmeter) movement with two coils** (current coil and pressure coil) on one spindle with **no control spring** — the pointer floats when idle and the deflection depends on the *ratio* V/I, i.e., directly on resistance, independent of crank speed.
- **Guard terminal** bypasses surface leakage so only volume insulation resistance is read.
- Used for **insulation resistance of cables, machines, transformers** (readings in MΩ).

6.12 AC Bridges — Overview

General four-arm bridge with impedances $Z_1Z_2Z_3Z_4$, AC source (usually 1 kHz oscillator), detector (headphones/tuned amplifier). Balance:

$Z_1 Z_4 = Z_2 Z_3$ — must be satisfied in **both magnitude and phase**, so two independent adjustments are needed.

Bridge	Measures	Balance equations	Remarks
Maxwell's inductance–capacitance bridge	**Medium-Q inductance (1 < Q < 10)** in terms of a standard capacitor	**$L_1 = R_2R_3C_4$**, $R_1 = R_2R_3/R_4$	Balance independent of frequency; standard capacitor (loss-free) preferred over standard inductor; unsuitable for high Q (R_4 becomes impractically large)
Hay's bridge	**High-Q inductance (Q > 10)**	**$L_1 = R_2R_3C_4 / (1 + \omega^2C_4^2R_4^2)$**; for Q > 10, $L_1 \approx R_2R_3C_4$	C_4 in **series** with R_4 (Maxwell has them in parallel); balance is **frequency-dependent**
Schering bridge	**Capacitance and dissipation factor (loss angle δ) of capacitors/insulation**, incl. HV testing	**$C_1 = C_2\cdot(R_4/R_3)$**, $R_1 = R_3\cdot(C_4/C_2)$; **$D = \tan\delta = \omega C_4R_4$**	The standard HV bridge for insulation/bushings/cables
Wien bridge	**Frequency** (also audio capacitance)	**$f = 1 / (2\pi\sqrt{(R_1R_2C_1C_2)})$**; with $R_1=R_2=R$, $C_1=C_2=C$: **$f = 1/(2\pi RC)$** and $R_4/R_3 = 2$	Basis of the **Wien-bridge oscillator**; frequency-sensitive
(Also asked) **Anderson bridge**	Inductance in terms of fixed capacitor (modification of Maxwell)	—	Very accurate over wide L range
(Also asked) **De Sauty bridge**	Capacitance (loss-free) comparison	$C_1/C_2 = R_4/R_3$	Simple, only for perfect capacitors

Wagner earthing device: removes errors due to stray capacitances to earth at the detector terminals.

6.13 Multimeters, DVMs and DMMs

Analog multimeter (VOM): one PMMC movement + switched shunts (DC A), multipliers (DC V), rectifier + multipliers (AC V), internal battery + series/shunt network (ohms — scale reads **right-to-left, non-linear**, zero-ohm adjust needed). Typical sensitivity 20 kΩ/V DC.

Digital Multimeter (DMM): signal conditioner (attenuator/amplifier, AC-DC converter, I→V shunt, ohms current source) + **A/D converter** + counter + display. Advantages: no parallax, high input impedance (≈10 MΩ constant on all ranges), auto-ranging, auto-polarity, better accuracy (±0.5 % to ±0.005 %), resolution specified in **digits (3½, 4½)** — a 3½-digit meter displays up to **1999** (max leading digit 1); **overranging** = the extra ½ digit.

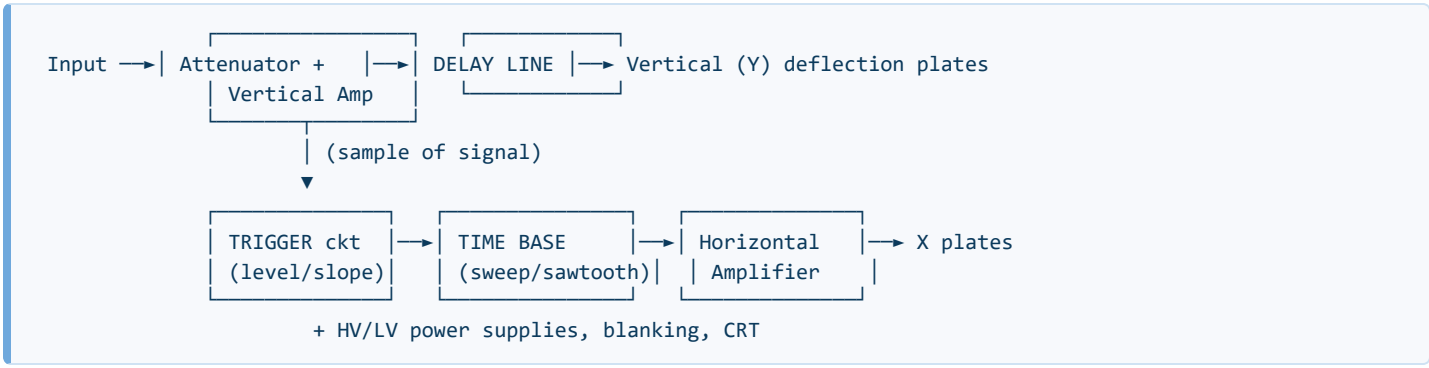
Digital voltmeter (DVM) types — one line each:

- **Ramp-type DVM:** measures the time a linear ramp takes to go from 0 to V_in; counts clock pulses in that gate time.
- **Staircase-ramp DVM:** DAC output steps up until it equals V_in; count = reading.
- **Integrating (voltage-to-frequency) DVM:** converts V to a proportional pulse rate; counts pulses — averages out noise.
- **Dual-slope integrating DVM:** integrates V_in for a fixed time, then de-integrates with a reference; reading independent of R, C and clock drift; **excellent 50 Hz noise rejection** — the standard DMM technique.
- **Successive-approximation DVM:** binary search using a DAC + comparator; **fastest** for given resolution.

6.14 Cathode Ray Oscilloscope (CRO) — In Depth ★★★

The CRO displays **voltage vs time** (or X vs Y) in real time; input impedance typically **1 MΩ ‖ 10–30 pF**.

6.14.1 Block Diagram



- **Vertical (Y) system:** input attenuator (V/div switch, compensated RC dividers) → wideband **vertical amplifier** (sets sensitivity & bandwidth) → delay line → Y-plates.
- **Trigger circuit:** picks a sample of the vertical signal (or EXT/LINE source), compares with the set **level** on the chosen **slope (+/−)**, and fires the time base at the same waveform point every sweep → **stable, stationary display**.
- **Time base (horizontal system):** generates a linear **sawtooth**: slow linear **sweep** (spot moves left→right at uniform speed = TIME/div) followed by rapid **retrace/flyback** (blanked). Modes: auto, normal, single. In X-Y mode the sawtooth is replaced by an external X signal.
- **Blanking:** negative gate to the control grid extinguishes the beam during retrace.

6.14.2 The Delay Line ★ (asked in 22-04-2025)

Why needed: The trigger + time-base circuits need **~80–200 ns** to start the sweep after the triggering edge arrives. Without compensation, the **leading edge of the very signal that caused the trigger would already be past** before the sweep starts, so the start of the waveform would never be seen. The **delay line, placed in the vertical channel after the point where the trigger signal**

is tapped, delays the signal to the Y-plates by slightly **more** than the trigger+sweep initiation time, so the sweep is already running when the signal edge reaches the plates → **the entire leading edge is displayed**.

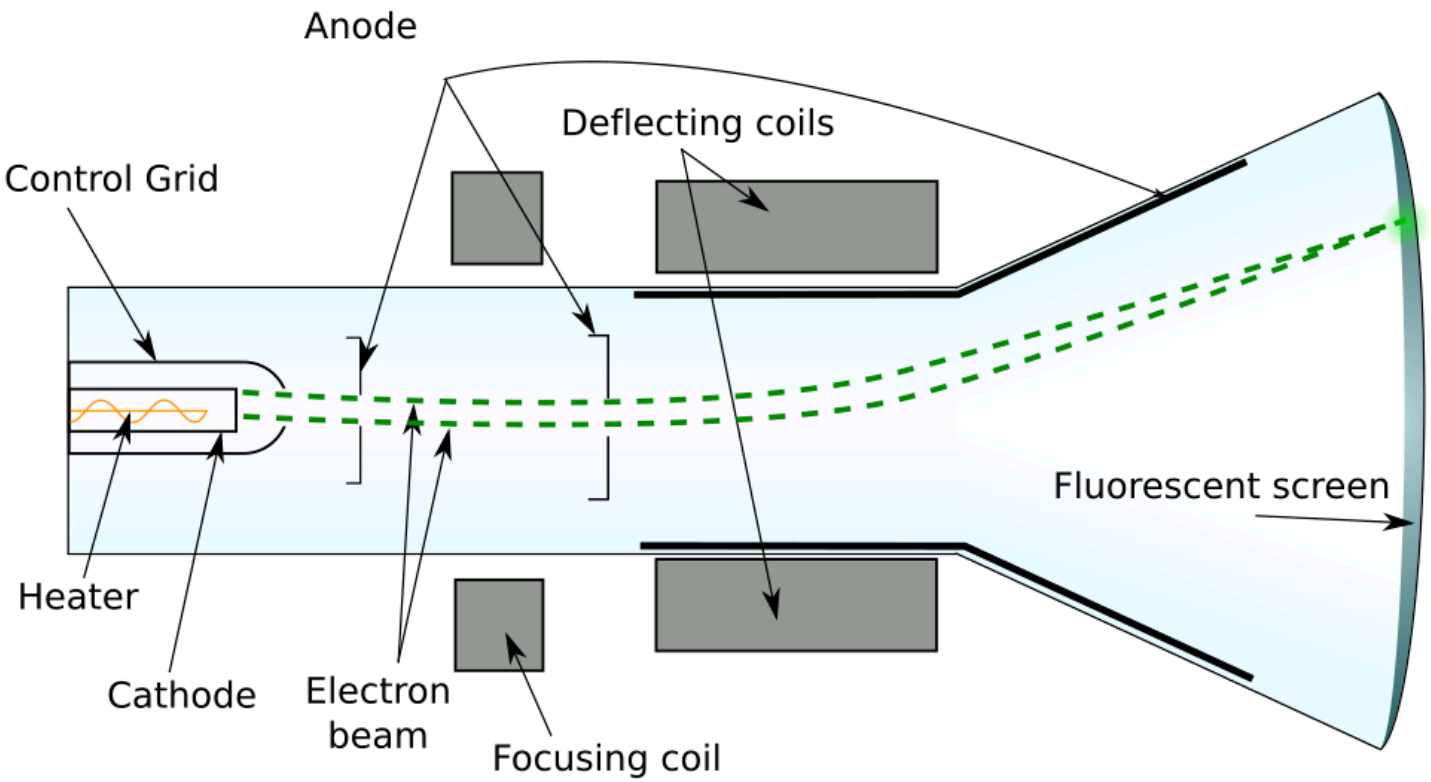
Two constructions:

1. **Lumped-parameter (electrical) delay line:** a cascade of n identical symmetrical **LC sections** (low-pass filter chain).

- Delay per section: **$t_s = \sqrt{(L \cdot C)}$**; total delay **$t_d = n \cdot \sqrt{(LC)}$**.
- Characteristic impedance **$Z_0 = \sqrt{(L/C)}$** — the line must be terminated in Z_0 to avoid reflections/ringing.
- Hence the **delay (effective electrical "length") is determined by the propagation speed of the signal along the line, which is fixed by the L and C values per section and the number of sections.** ⭐ *This is exactly the 2025 PYQ.*

1. **Distributed-parameter delay line:** a special helical-inner-conductor **coaxial cable**; delay ≈ 180 ns/m typical (vs ~5 ns/m for ordinary coax); used in high-bandwidth scopes.

6.14.3 CRT Construction



Cathode-ray tube of a CRO.

Evacuated funnel-shaped glass tube containing:

1. **Electron gun:** indirectly heated **cathode** (barium/strontium oxide coated) emits electrons → **control grid** (negative w.r.t. cathode; sets beam current ⇒ **intensity/brightness** control) → **pre-accelerating & accelerating anodes** (high +ve) → **focusing anode** (electrostatic lens; **focus** control). "Astigmatism" control shapes the spot round.
2. **Deflection system:** two pairs of **electrostatic deflection plates** — vertical (Y) plates deflect vertically, horizontal (X) plates deflect horizontally. (Electrostatic deflection is used in CROs because it responds to high frequencies; TVs used magnetic deflection.)

- **Deflection on screen:** $D = (l \cdot L \cdot V_d) / (2 \cdot d \cdot V_a)$

where l = plate length, L = plate-centre-to-screen distance, d = plate separation, V_d = deflecting voltage, V_a = accelerating voltage.

- **Deflection sensitivity S = D/V_d = lL/(2dV_a)** (mm/V) — **inversely proportional to accelerating voltage**; **deflection factor = 1/S (V/mm)**.
- **Post-deflection acceleration (PDA)** electrodes brighten the trace without hurting sensitivity.

1. **Screen:** inside face coated with **phosphor** (e.g., P31 for general use — green, medium persistence; P11 short persistence for photography; long-persistence types for slow phenomena). Light emission on impact = **fluorescence**; continued glow after = **phosphorescence**; **persistence** = time the glow lasts.
2. **Aquadag coating:** conductive **graphite coating** on the inner funnel; it (i) is held at accelerating potential to accelerate/collect **secondary-emission electrons** from the screen, completing the beam return path, and (ii) shields the beam.

6.14.4 Probes ☆ (asked in 22-04-2025)

Probe	Description
1:1 (direct) probe	Shielded coaxial lead, no attenuation; loads the circuit with full cable + input capacitance (~100 pF); only for low-frequency, strong signals
10:1 attenuating (passive) probe	9 MΩ resistor **with adjustable parallel capacitor** in the tip, working against the scope's 1 MΩ ‖ C_in → attenuation 10×, input resistance raised to **10 MΩ**, capacitive loading cut ~10×
Probe compensation	The tip capacitor is trimmed so $\mathbf{R_1C_1 = R_2C_2}$ (a *compensated attenuator* — division ratio then independent of frequency). Checked on the scope's ~1 kHz **CAL square wave**: flat top = compensated; rounded = under-compensated; overshoot/spiked = over-compensated
Active (FET) probe	FET buffer at the tip: very low capacitance, no attenuation, for high frequency
Current probe	Clamp-on Hall/transformer sensor — measures current without breaking circuit

Why is the probe cable shielded (coaxial)? ☆ The signal conductor is surrounded by a **grounded braided shield**. The shield intercepts external electric fields and diverts induced currents to ground, so **stray pickup of external electrical interference — mains hum, RF, switching noise (EMI) — is prevented from corrupting the small measured signal**. It also defines a controlled impedance and keeps the probe from radiating. *This is exactly the 2025 PYQ.*

6.14.5 Measurements with a CRO

- **Amplitude:** V = (vertical divisions) × (VOLTS/div) × probe factor. Peak-to-peak read directly; V_rms = V_pp/(2√2) for a sine.
- **Time period:** T = (horizontal divisions for one cycle) × (TIME/div).
- **Frequency:** f = 1/T.
- **Phase difference (dual trace):** φ = (Δt / T) × 360°.
- **Phase by ellipse (X-Y):** sin φ = (Y-intercept)/(Y-max) = B/A.
- **Frequency by Lissajous figures (X-Y mode):**

f_Y / f_X = (number of horizontal tangencies) / (number of vertical tangencies)

(= points of tangency to a horizontal line / to a vertical line)

Ratio f_y : f_x	Pattern (equal amplitudes)
1 : 1, φ = 0°	Straight line at 45°
1 : 1, φ = 90°	Circle
1 : 1, 0<φ<90°	Ellipse
1 : 1, φ = 180°	Straight line at 135°
2 : 1	Figure-of-eight (∞ rotated) — 2 tangencies top, 1 on side
3 : 1	Three-lobed pattern
3 : 2	Complex loop — 3 horizontal tangencies, 2 vertical

6.14.6 Dual-Trace vs Dual-Beam; DSO

- **Dual-trace CRO: one** electron beam, electronically switched between two Y channels — **ALTERNATE mode** (a full sweep of each channel alternately; good for fast signals) and **CHOP mode** (switches between channels at ~100–500 kHz during each sweep; good for slow signals). Cannot truly capture two one-shot events simultaneously.
- **Dual-beam CRO: two** independent beams (two guns or split beam) with separate vertical systems — genuinely simultaneous display; costlier.
- **Digital Storage Oscilloscope (DSO):** input → attenuator/amp → **sample-and-hold → ADC → acquisition memory → microprocessor → display** (raster LCD).

Advantages: stores waveforms indefinitely; captures **single-shot/transient** events; **pre-trigger viewing** (see what happened *before* the trigger); automatic measurements, cursors, FFT, averaging (noise reduction), storage/USB/printing; no CRT drift.

Key specs: **sampling rate (Sa/s ≥ 2× bandwidth by Nyquist, practically ≥10×)**, record length, resolution (8-bit typical). Modes: real-time sampling (single shot) and equivalent-time sampling (repetitive signals). *Aliasing* occurs if sampling is too slow.

6.15 Display Panels — LED and LCD

6.15.1 LED Displays

- **LED** = forward-biased **p–n junction** in a direct-band-gap material (GaAs — IR; GaAsP — red/orange/yellow; GaP, GaN/InGaN — green/blue). Emits light by **injection electroluminescence — radiative recombination** of electrons and holes; photon energy ≈ E_g, so **λ = 1240/E_g(eV) nm**.
- Typical drop 1.8–3.3 V; always used with a **current-limiting resistor**; fast (ns), long life, rugged, wide temperature range.
- **Seven-segment display:** segments **a–g (+ dp)**.
- **Common-anode (CA):** all anodes tied to +V; a segment lights when its cathode is pulled **LOW** (driver sinks current, e.g., 7447 decoder).
- **Common-cathode (CC):** all cathodes grounded; segment lights on **HIGH** (e.g., 7448/4511 decoder).
- **Multiplexing:** in a multi-digit display all like segments are bussed together and digits are enabled **one at a time in rapid rotation (> ~50–100 Hz refresh)**; persistence of vision shows all digits lit. Saves driver pins and power.
- Dot-matrix (5×7) LED displays show alphanumerics.

6.15.2 LCD Displays

- **Principle:** liquid crystals (nematic phase) do not emit light — they **modulate external light**. The dominant type is the **Twisted Nematic (TN) field-effect** cell: LC fluid between two glass plates with transparent **ITO electrodes** and crossed **polarizers**; the molecular director twists 90° across the cell.
- **No field:** the twist rotates the polarization 90° → light passes the second polarizer → segment **clear/bright**.

- **Field applied (few volts AC):** molecules align with the field, twist destroyed, no rotation → light blocked → segment appears **dark**.
- **Driven by AC (30–100 Hz square wave), never DC** — DC causes electrochemical degradation of the electrodes.
- **Dynamic-scattering** LCDs (older) become milky-white under field.
- Reflective (uses ambient light + rear mirror) or **backlit/transmissive**.

6.15.3 LED vs LCD Comparison ☆

Feature	LED display	LCD display
Light	**Emits** its own light (active)	**Modulates** ambient/backlight (passive)
Power	mA per segment (10–20 mA) — relatively high	**µW per segment** — extremely low (ideal for watches, DMMs, calculators)
Drive	DC, current-driven	**Low-voltage AC**, voltage-driven
Visibility	Excellent in dark; washes out in sunlight	Excellent in bright light; needs backlight in dark
Viewing angle	Wide	Limited (TN)
Speed	ns — very fast	**ms — slow**; sluggish at low temperature
Temperature range	Wide (−40 to +85 °C)	Narrow (LC freezes/clears outside range)
Life/ruggedness	Very long, rugged	Long but fluid-based, fragile glass

6.15.4 OLED (brief)

Organic LED: thin films of organic semiconductor emit light on current injection — **self-emissive** (no backlight), so true blacks, very high contrast, wide viewing angle, thin/flexible panels, fast response; drawbacks: differential ageing of blue emitters and burn-in. Each pixel is its own light source — conceptually "an LED display with LCD-class resolution."

6.16 Transducers — In Depth ☆☆☆

A **transducer** converts one form of energy/physical quantity into another — in instrumentation, usually a non-electrical measurand into an **electrical** signal. (Sensor = sensing element; transducer = sensor + conversion.)

6.16.1 Classification

Basis	Types	Examples
Power source	**Active (self-generating)** — generate their own electrical output, need **no external supply**	Thermocouple, piezoelectric, photovoltaic cell, electromagnetic (tachogenerator)
	Passive — need external excitation; measurand changes a passive parameter (R, L, C)	Strain gauge, RTD, thermistor, LVDT, capacitive, LDR
Conversion stage	**Primary** — first element sensing the measurand (may give mechanical output)	Bourdon tube, diaphragm, bimetal
	Secondary — converts the primary's output to electrical	LVDT sensing Bourdon-tube tip motion
Output nature	**Analog** (continuous)	LVDT, thermocouple, strain gauge
	Digital (discrete/pulse)	Shaft encoder, turbine flowmeter pulses
Principle	Resistive, inductive, capacitive, piezoelectric, thermoelectric, photoelectric, Hall-effect	see below

Selection criteria: range & span, required accuracy/linearity/sensitivity, operating environment (temperature, corrosion, vibration), loading effect on the measurand, frequency response/speed, output impedance & signal level, ruggedness, size, cost, availability of excitation.

6.16.2 Resistive Transducers

(a) Potentiometer (POT): resistance wire/film with sliding wiper; **linear or rotary displacement → resistance/voltage ratio**. Simple, cheap, high output; but friction, wear, limited resolution (wire-wound), loading error if output is loaded.

(b) Strain gauge ⭐: a metal foil/wire grid bonded to the specimen; strain changes its resistance.

Gauge factor:

$G_f = (\Delta R/R) / (\Delta l/l) = (\Delta R/R)/\varepsilon = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon$

(1 from length, 2ν from area/Poisson effect ν ≈ 0.3, last term = piezoresistive effect.)

- Metal gauges: $G_f \approx$ **2** (constantan/advance, nichrome); semiconductor gauges: $G_f \approx$ **100–200** (large piezoresistive term) but temperature-sensitive and non-linear.
- Used in a **Wheatstone bridge** (quarter/half/full) — full bridge with 4 active gauges gives 4× output and temperature compensation; **dummy gauge** compensates temperature in quarter-bridge.

> **Worked Example 6.8 (strain gauge).**
>
> A 120 Ω strain gauge, $G_f = 2.0$, is bonded to a steel member under stress $\sigma = 100$ MN/m² ($E = 200$ GN/m²). Find ΔR.
>
> Strain $\varepsilon = \sigma/E = 100\times10^{6} / 200\times10^{9} = 5\times10^{-4}$ (= 500 microstrain).
>
> $\Delta R = G_f \cdot \varepsilon \cdot R = 2 \times 5\times10^{-4} \times 120 =$ **0.12 Ω** ($\Delta R/R = 0.1$ %).

(c) RTD (Resistance Temperature Detector): pure metal (usually **platinum**) whose resistance rises almost linearly with temperature — **positive temperature coefficient**.

$R_t = R_0(1 + \alpha\Delta T)$, $\alpha_{Pt} \approx 0.00385$ /°C.

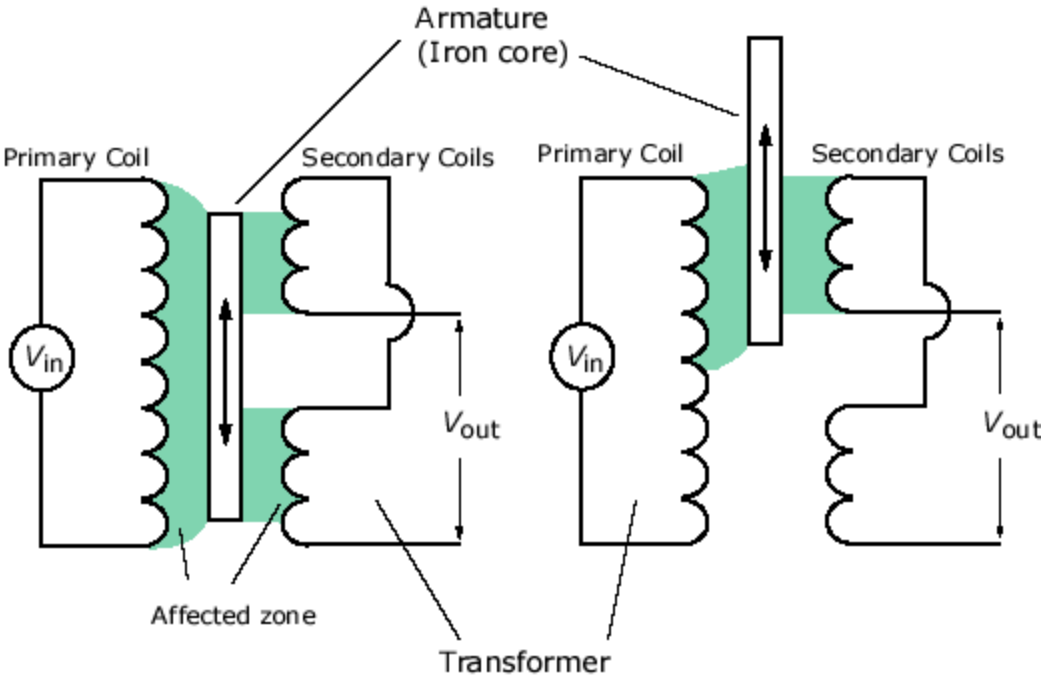
PT100 = platinum RTD with R = 100 Ω at 0 °C (≈ 138.5 Ω at 100 °C). Range ≈ −200 to +850 °C; excellent accuracy & stability; 3-wire/4-wire connection cancels lead resistance; needs excitation current (self-heating error must be limited).

(d) Thermistor: sintered metal-oxide semiconductor bead; **NTC** — resistance **falls exponentially** with temperature:

$R_T = R_0 \cdot \exp[\beta(1/T - 1/T_0)]$, β ≈ 3000–4500 K.

Very high sensitivity (−4 %/°C, ~10× RTD), small, fast, cheap; but **highly non-linear**, narrow range (−50 to +300 °C), interchangeability issues. (PTC thermistors exist — used as resettable fuses/protection, not measurement.)

6.16.3 Inductive — LVDT ☆☆ (exam favourite)



LVDT displacement transducer.

Linear Variable Differential Transformer — passive, inductive **displacement** transducer.

Construction: one **primary** winding centred between **two identical secondaries wound in series opposition**, on a hollow former; a **soft-iron (Ni-Fe) core** attached to the moving member slides inside without touching.

Working: primary excited by AC (50 Hz–20 kHz).

- Core central (**null position**): equal coupling → V_s1 = V_s2 → net output **$V_0 = V_{s1} - V_{s2} = 0$**.
- Core moves toward secondary 1: V_s1 > V_s2 → output rises **in phase** with a reference.
- Core moves the other way: output rises with **180° phase reversal**.

Thus $|V_0|$ ∝ displacement (linear over the rated range) and **phase gives direction** (phase-sensitive detector converts to signed DC).

Residual (null) voltage: the small output (< 1 % of FS) that remains at the null position — caused by harmonics in excitation, stray capacitance and winding asymmetry. It limits resolution; minimized by good construction and phase-sensitive demodulation.

Advantages: frictionless & wear-free (no contact between core and coils) → essentially **infinite resolution** and very long life; excellent **linearity** (~0.5 %); high sensitivity/output; rugged; overload-safe; DC-out versions available.

Limitations: sensitive to stray AC fields (needs shielding), core mass limits fast dynamics, temperature effects, needs AC excitation + demodulation.

Applications: displacement 1 μm – 25 cm; secondary transducer for pressure (with Bourdon/diaphragm), load, thickness, machine-tool gauging. (**RVDT** = rotary version for angles ±40°.)

6.16.4 Capacitive Transducers

$C = \varepsilon_0 \varepsilon_r A/d$. Vary any of the three:

- **Δd (gap change):** most sensitive, non-linear ($C \propto 1/d$) — pressure microphones, diaphragm pressure sensors; differential (push-pull) arrangement linearizes.
- **ΔA (area overlap):** linear — angular/linear displacement.
- **Δε_r (dielectric change):** liquid **level** gauging, moisture/humidity sensors.

High impedance output → needs shielded leads and AC bridge/oscillator circuits; very sensitive, tiny forces needed, good frequency response; affected by stray capacitance.

6.16.5 Piezoelectric Transducers

Active transducer: mechanical stress on an asymmetric crystal displaces its internal charge centres → **charge appears on the faces** (piezoelectric effect); converse effect used in ultrasonic generators.

- Materials: **quartz** (stable, natural), Rochelle salt (most sensitive, fragile), synthetic ceramics **barium titanate, PZT** (lead-zirconate-titanate).
- Charge output **Q = d·F** (d = charge sensitivity, C/N; for quartz d ≈ 2.3 pC/N); voltage V = Q/C = g·t·P.
- Very high output impedance → **charge amplifier** needed; **cannot measure true static (DC) quantities** — charge leaks away; superb for **dynamic** measurements: vibration **accelerometers**, force/pressure transients, ultrasonics, spark igniters.

6.16.6 Thermocouple ☆

Seebeck effect: in a circuit of **two dissimilar metals** with junctions at different temperatures, an EMF proportional (approx.) to the junction temperature difference is generated: **$E \approx \alpha(T_{hot} - T_{cold})$**. Active transducer.

(Related: **Peltier effect** — heat absorbed/liberated at a junction carrying current; **Thomson effect** — EMF along a single conductor with a temperature gradient.)

Type	Materials (+/−)	Range (approx.)	Sensitivity	Notes
J	Iron / Constantan	−40 to 750 °C	~55 μV/°C	Cheap; iron rusts, not for oxidizing high temp
K	Chromel / Alumel	−200 to 1260 °C	~41 μV/°C	**General-purpose workhorse**, most common
T	Copper / Constantan	−200 to 350 °C	~43 μV/°C	Best for low/cryogenic temperatures
E	Chromel / Constantan	−200 to 900 °C	~68 μV/°C	**Highest sensitivity**
R	Pt–13 %Rh / Pt	0 to 1600 °C	~10 μV/°C	High temperature, precious metal
S	Pt–10 %Rh / Pt	0 to 1600 °C	~10 μV/°C	Standard instrument thermocouple (old IPTS standard)
B	Pt-30%Rh / Pt-6%Rh	up to 1800 °C	very low	Highest range

Cold-junction compensation (CJC): thermocouple EMF depends on **both** junction temperatures; tables assume the reference junction at **0 °C**. In practice the reference sits at ambient, so either keep it in an ice bath, or (modern way) **measure the terminal-**

block temperature with an RTD/thermistor/IC sensor and add the corresponding EMF electronically. **Law of intermediate metals** allows the meter's copper terminals in the loop without error if both connections are at the same temperature; **compensating/extension cables** must match the thermocouple alloys.

6.16.7 Photoelectric Transducers

- **Photodiode:** reverse-biased p-n/PIN junction; light generates carriers → reverse (photo)current ∝ **illumination**; very fast (ns) — optical communication, encoders, light meters. (Phototransistor = photodiode + gain, slower.)
- **LDR (photoconductive cell, CdS/CdSe):** resistance **falls** (MΩ → hundreds of Ω) with illumination; cheap, sensitive, but slow (ms–s) — light-operated relays, street-light controls.
- **Photovoltaic (solar) cell: active** — illuminated junction generates EMF (~0.5 V Si) with no bias; power source + lux meters.
- (Photoemissive tubes/photomultipliers: electron emission from a photocathode; PMT for extremely low light.)

6.16.8 Hall-Effect Sensor

Current-carrying semiconductor slab in a **perpendicular magnetic field B** develops a transverse voltage:

$V_H = (B \cdot I)/(n \cdot q \cdot t) = R_H \cdot B \cdot I/t$ (t = thickness, R_H = Hall coefficient, large in semiconductors).

Measures **magnetic field**, and hence anything convertible to B: **contactless current sensing (clamp meters), position/proximity, brushless-DC rotor position, keyboards, speed sensing**. Output exists even for **DC fields** (unlike inductive pickups); polarity of V_H identifies carrier type (n or p).

6.16.9 Load Cell

Elastic member (column, ring, cantilever or shear beam) + **four bonded strain gauges in a full Wheatstone bridge** → output mV/V proportional to applied **force/weight**; temperature-compensated, linear to 0.02 %; the heart of electronic weighing machines, tank weighing, and material-testing machines. (Hydraulic/pneumatic load cells exist for special environments; piezoelectric washers for dynamic force.)

6.16.10 Digital Encoders (one paragraph)

A **shaft encoder** converts angular (or, as a linear scale, translational) position directly into digital form. **Incremental encoders** output pulse trains from an optical disc (LED–photodiode reading slots); position = pulse count, speed = pulse rate, and two channels in **quadrature (90°)** give direction; an index pulse marks zero — but position is lost at power-down. **Absolute encoders** carry a multi-track coded disc that outputs a unique word per position; **Gray code** is used instead of straight binary so that **only one bit changes between adjacent positions**, preventing gross misreads at track boundaries. Resolution = $360°/2^n$ for n tracks; encoders are noise-immune, wear-free (optical) and interface directly to digital systems.

6.17 Worked Numericals (Consolidated Set)

(Examples 6.1–6.8 above are part of the count; three more below complete the mandated set.)

> **Worked Example 6.9 (CRO time-base reading).**

> A CRO displays 2.5 complete cycles across 10 horizontal divisions with TIME/div = 20 µs. The waveform is 6 divisions peak-to-peak with VOLTS/div = 0.5 V via a 10:1 probe. Find f and V_{rms} (sine).

> One cycle occupies 10/2.5 = 4 div ⟹ T = 4 × 20 µs = 80 µs ⟹ **f = 12.5 kHz**.

> V_{pp} = 6 × 0.5 × 10 = 30 V ⟹ V_m = 15 V ⟹ **$V_{rms} = 15/\sqrt{2} = 10.6$ V**.

Worked Example 6.10 (Lissajous frequency ratio).

An unknown frequency f_y drives the Y-input and a 2 kHz standard drives X. The stationary pattern touches a horizontal tangent line at **3** points and a vertical tangent line at **2** points. Find f_y.

f_y/f_x = horizontal tangencies / vertical tangencies = 3/2 $\Longrightarrow$ f_y = (3/2)(2 kHz) = **3 kHz**.

Worked Example 6.11 (delay line).

A CRO lumped delay line has 30 identical sections with L = 60 μH and C = 1.5 pF per section. Find the delay per section, total delay and characteristic impedance.

$t_s = \sqrt{(LC)} = \sqrt{(60\times10^{-6} \times 1.5\times10^{-12})} = \sqrt{(9\times10^{-17})} = 9.49$ ns? → $\sqrt{(9.0\times10^{-17})} = 9.49\times10^{-9}$ s ≈ **9.5 ns/section**? Recheck: $60\times10^{-6} \times 1.5\times10^{-12} = 90\times10^{-18} = 9.0\times10^{-17} \Longrightarrow \sqrt{} = 9.49\times10^{-9}$ s ✓.

Hmm — that is large per section; realistic lines use smaller L. Taking the given values: t_s ≈ **9.5 ns**, total t_d = 30 × 9.5 ns ≈ **285 ns**; $Z_0 = \sqrt{(L/C)} = \sqrt{(60\times10^{-6}/1.5\times10^{-12})} = \sqrt{(4\times10^{7})}$ = **6.32 kΩ** — the line must be terminated in ~6.3 kΩ to prevent reflections.

Takeaway: delay is set purely by **n√(LC)** — the 2025 PYQ point.

Worked Example 6.12 (deflection sensitivity).

In a CRT, plate length l = 2 cm, plate separation d = 0.5 cm, plate-centre-to-screen distance L = 20 cm, accelerating voltage V_a = 2000 V. Find deflection sensitivity.

$S = lL/(2dV_a) = (0.02 \times 0.20)/(2 \times 0.005 \times 2000) = 0.004/20 = 2\times10^{-4}$ m/V = **0.2 mm/V**. Deflection factor = 1/S = **5 V/mm**.

(Count of worked numericals in this chapter: 6.1 limiting error, 6.2 % error, 6.3 energy meter, 6.4 & 6.5 shunt, 6.6 multiplier, 6.7 loading, 6.8 strain gauge, 6.9 CRO reading, 6.10 Lissajous, 6.11 delay line, 6.12 deflection — 12 total.)

6.18 PYQ Corner — Fully Solved

PYQ 1 (RRB JE CBT-2, 22-04-2025)

Q. Why is the cable of an oscilloscope probe shielded?

(a) To increase its mechanical strength (b) To reduce pickup of external electrical interference/noise (c) To increase the probe's attenuation ratio (d) To provide the delay required by the time base

Solution. The probe carries a small, high-impedance signal from the circuit to the CRO. An unshielded lead acts as an antenna, picking up mains hum, RF and switching noise through capacitive and electromagnetic coupling. The **grounded coaxial shield** surrounds the inner signal conductor, intercepts external fields and returns induced currents to ground, so the displayed signal is not corrupted. **Answer: (b).** (Note: shielding does not set attenuation — the tip RC divider does — and delay is the job of the delay line inside the scope.)

PYQ 2 (RRB JE CBT-2, 22-04-2025)

Q. The length (delay) of the electrical delay line in a CRO is determined by:

(a) The brightness of the trace (b) The speed at which the signal propagates along it, fixed by its L and C parameters (c) The phosphor persistence (d) The vertical amplifier gain

Solution. The electrical delay line is a lumped low-pass chain of identical LC sections. A signal travels along it at a finite speed set by L and C; delay per section = **$\sqrt{LC}$** and total delay = **$n\sqrt{LC}$**, with $Z_0 = \sqrt{L/C}$. The physical/electrical "length" needed is therefore governed by the **propagation speed of the signal on the line** — i.e., by its LC parameters — so that the vertical signal reaches the plates ~100–200 ns after the trigger has started the sweep. **Answer: (b).**

PYQ-style 3

Q. A 1 mA, 100 Ω movement is to be converted into a 0–10 A ammeter. The required shunt is:

(a) 0.01 Ω (b) 0.1 Ω (c) 0.010001 Ω (d) 1 Ω

Solution. m = 10/0.001 = 10 000. $R_{sh} = R_m/(m-1) = 100/9999$ = **0.010001 Ω ≈ 0.01 Ω**. Strictly **(c)** (≈ (a) to 2 s.f.); in exam options the exact value 100/9999 Ω is the correct pick. **Answer: (c).**

PYQ-style 4

Q. In an LVDT, when the core is exactly at the centre, the output voltage is ideally zero, but a small voltage is still observed. This is called:

(a) Offset voltage (b) Residual/null voltage (c) Ripple voltage (d) Quadrature drive

Solution. With the two secondaries in series opposition, perfect symmetry would give zero at centre; in practice harmonics, stray capacitance and winding mismatch leave a small **residual (null) voltage**, typically < 1 % of full-scale output. **Answer: (b).**

PYQ-style 5

Q. A Wheatstone bridge has P = 1000 Ω, Q = 100 Ω, and balances with the standard arm S = 46.3 Ω. The unknown resistance is:

(a) 4.63 Ω (b) 46.3 Ω (c) 463 Ω (d) 4630 Ω

Solution. At balance $R = (P/Q)\cdot S = (1000/100) \times 46.3$ = **463 Ω**. **Answer: (c).**

6.19 Practice MCQs (25) with Answer Key

1. The first element of a generalized measurement system that receives energy from the measurand is the:

(a) signal conditioner (b) primary sensing element (c) data presentation element (d) transmission channel

2. An instrument gives readings 50.1, 50.0, 50.2, 50.1 V when the true value is 49.0 V. The instrument is:

(a) accurate and precise (b) accurate but not precise (c) precise but not accurate (d) neither

3. Parallax error is a:

(a) random error (b) gross/observational error (c) instrumental systematic error (d) environmental error

4. A 0–100 V voltmeter guaranteed to ±2 % of FSD reads 40 V. The possible % error in the reading is:

(a) ±2 % (b) ±0.8 % (c) ±5 % (d) ±8 %

5. Gravity control gives a scale that is:

(a) uniform (b) cramped at the lower end (c) cramped at the upper end (d) logarithmic

6. Eddy-current damping cannot be used in moving-iron instruments because:

(a) it is too weak (b) the required permanent magnet would distort the operating field (c) MI instruments need no damping (d) the vane is non-conducting

7. A PMMC meter connected to 50 Hz AC will:

(a) read RMS (b) read average (c) read peak (d) show zero/vibrate about zero

8. In a PMMC instrument, the deflecting torque is:

(a) $\propto I^2$ (b) $\propto I$ (c) $\propto \sqrt{I}$ (d) $\propto 1/I$

9. An energy meter with only the potential coil energized rotates slowly. This is prevented by:

(a) a lag coil (b) two holes drilled in the disc (c) moving the brake magnet inward (d) shading the series magnet

10. A dynamometer wattmeter deflection is proportional to:

(a) $V \cdot I$ (b) $V \cdot I \cdot \cos\varphi$ (c) $V \cdot I \cdot \sin\varphi$ (d) I^2

11. A 50 μA, 2 kΩ movement is used as a voltmeter. Its sensitivity is:

(a) 2 kΩ/V (b) 10 kΩ/V (c) 20 kΩ/V (d) 40 kΩ/V

12. For the meter of Q.11 to read 0–100 V, the multiplier resistance is:

(a) 2 MΩ (b) 1.998 MΩ (c) 2.002 MΩ (d) 200 kΩ

13. The secondary of a current transformer should never be open-circuited because:

(a) the primary fuse blows (b) dangerously high voltage is induced and the core saturates/overheats (c) the secondary current becomes excessive (d) the burden rises to infinity

14. Kelvin double bridge is preferred for low resistances because it eliminates the effect of:

(a) supply variations (b) contact and lead resistances (c) temperature (d) galvanometer resistance

15. Insulation resistance of a cable is measured with:

(a) Kelvin bridge (b) Wheatstone bridge (c) Megger (d) potentiometer

16. The bridge best suited for measuring an inductor with $Q > 10$ is:

(a) Maxwell (b) Hay (c) Schering (d) De Sauty

17. The Schering bridge is chiefly used to measure:

(a) frequency (b) low resistance (c) capacitance and loss angle of insulation (d) mutual inductance

18. A Wien bridge with $R_1 = R_2 = 10$ kΩ and $C_1 = C_2 = 0.01$ μF balances at a frequency of about:

(a) 15.9 Hz (b) 159 Hz (c) 1.59 kHz (d) 15.9 kHz

19. A 3½-digit DVM on its 20 V range has a resolution of:

(a) 1 mV (b) 10 mV (c) 100 mV (d) 1 V

20. In a CRO, the delay line is placed:

(a) before the attenuator (b) in the trigger circuit (c) in the vertical channel between amplifier and Y-plates (d) after the time base

21. The aquadag coating in a CRT serves to:

(a) focus the beam (b) collect secondary-emission electrons and complete the return path (c) generate the sweep (d) blank the retrace

22. A 10:1 probe is "compensated" when:

(a) its resistance equals the scope input resistance (b) $R_1C_1 = R_2C_2$ so division is frequency-independent (c) the cable is exactly 1 m (d) the shield is left floating

23. A Lissajous pattern shows 4 tangencies to a horizontal line and 1 to a vertical line. If f_x = 500 Hz, f_y is:

(a) 125 Hz (b) 500 Hz (c) 1 kHz (d) 2 kHz

24. Which transducer is active (self-generating)?

(a) strain gauge (b) LVDT (c) thermocouple (d) thermistor

25. PT100 denotes:

(a) a thermistor of 100 Ω at 25 °C (b) a platinum RTD of 100 Ω at 0 °C (c) a thermocouple with 100 µV/°C (d) a potentiometer of 100 Ω

Answer Key with Brief Explanations

Q	Ans	Why
1	b	Primary sensing element (detector) contacts the measurand first
2	c	Tight cluster (precise) but offset ~1.1 V from truth (inaccurate)
3	b	Human/observational mistake; mirror scales reduce it
4	c	±2 V absolute; 2/40 × 100 = ±5 %
5	b	$T_c = WI \sin\theta \rightarrow$ small controlling torque at small θ → crowded low end
6	b	A damping magnet would superpose on the operating flux of the MI coil
7	d	Mean torque zero on AC; pointer merely vibrates
8	b	$T_d = NBAI = GI \rightarrow$ linear scale
9	b	Creep; holes rest under the shunt-magnet pole
10	b	Average torque ∝ VI cos φ = true power
11	c	$S = 1/I_{fsd} = 1/50\ \mu A = 20\,000\ \Omega/V$
12	b	$R_{total} = 20\ k\Omega/V \times 100\ V = 2\ M\Omega$; $R_s = 2\ M\Omega - 2\ k\Omega = 1.998\ M\Omega$
13	b	No opposing MMF → flux soars → high secondary EMF + core heating
14	b	Second ratio-arm pair cancels yoke/lead resistance
15	c	Megger: hand-driven DC generator + ratiometer ohmmeter
16	b	Hay (series R-C arm) suits high-Q; Maxwell suits 1<Q<10
17	c	Standard bridge for capacitor/insulation tan δ, incl. HV
18	c	$f = 1/(2\pi RC) = 1/(2\pi \times 10^4 \times 10^{-8}) = 1591$ Hz
19	b	3½ digits → 2000 counts; 20 V/2000 = 10 mV
20	c	Delays signal so the sweep (already triggered) shows the leading edge
21	b	Graphite layer at anode potential collects secondaries, shields beam
22	b	Compensated attenuator condition; check on CAL square wave
23	d	$f_y/f_x = 4/1 \rightarrow f_y = 4 \times 500 = 2000$ Hz

Q	Ans	Why
24	c	Seebeck EMF is self-generated; others need excitation
25	b	Platinum, 100 Ω at 0 °C, α ≈ 0.00385/°C

6.20 Rapid-Revision Sheet

Torque & Instrument Formulas

Item	Formula / fact				
PMMC torque	$T_d = NBAI = GI$; $\theta \propto I$ (uniform scale, DC only, eddy-current damping)				
Moving iron	$T_d = \frac{1}{2} I^2 dL/d\theta$; $\theta \propto I^2$, AC+DC (RMS), air-friction damping				
Dynamometer	$T \propto i_1 i_2 dM/d\theta$; wattmeter reads $VI \cos\varphi$, uniform scale				
Energy meter	$N \propto$ power; revolutions $\propto$ energy; creep → 2 holes in disc; brake magnet in → faster				
Spring control	$T_c = K\theta$	Gravity control	$T_c = WI \sin\theta$ (vertical use, cramped scale)		
Shunt	**$R_{sh} = R_m/(m-1)$**, $m = I/I_m$				
Multiplier	**$R_s = R_m(m-1)$**, $m = V/V_m$				
Sensitivity	$S = 1/I_{fsd}$ (Ω/V); $R_v = S \times$ range				
CT/PT	CT primary in series (sec. 5 A/1 A); PT across line (sec. 110 V); **never open a live CT secondary**; burden in VA				
Wheatstone	$R = (P/Q)S$	Kelvin double	low R, cancels lead/contact R	Megger	insulation MΩ
CRT sensitivity	$S = lL/(2dV_a)$ — ↓ as V_a ↑				
Delay line	$t_d = n\sqrt{(LC)}$; $Z_0 = \sqrt{(L/C)}$; shielded probe → noise immunity				
Lissajous	f_y/f_x = horizontal tangencies / vertical tangencies				
Limiting error	worst at low scale → use upper third				

AC Bridge Table

Bridge	Measures	Key relation
Maxwell	L (1 < Q < 10)	$L = R_2R_3C_4$ (freq-independent)
Hay	L (Q > 10)	$L \approx R_2R_3C_4$ (freq-dependent)
Schering	C & tan δ (insulation, HV)	$C_1 = C_2R_4/R_3$; $D = \omega C_4R_4$
Wien	Frequency	$f = 1/(2\pi RC)$, $R_4/R_3 = 2$
Kelvin (DC)	Low R	$R = (P/Q)S$
De Sauty	Loss-free C	$C_1/C_2 = R_4/R_3$

Transducer Classification Table

Class	Members	Memory hook
Active (no excitation)	Thermocouple, piezoelectric, photovoltaic, tachogenerator	"They **generate**"
Passive (R, L, C change)	Strain gauge, RTD, thermistor, POT (R); LVDT (L); capacitive (C); LDR	"They **modulate**"
Resistive facts	G_f = 1+2ν+piezoresistive ≈ **2** (metal), 100–200 (semiconductor); **PT100** = Pt, 100 Ω @ 0 °C, **+TC**; thermistor **NTC**, exponential	
LVDT	Two secondaries in series opposition; null output at centre; phase reverses through null; **residual voltage** at null; frictionless, infinite resolution	
Piezoelectric	Active; quartz/PZT; Q = dF; **dynamic only** (no static DC)	
Hall	V_H = BI/(nqt); contactless current & position sensing	
Load cell	4 strain gauges, full bridge	
Encoders	Incremental (quadrature, count) vs absolute (**Gray code**, unique word)	

Thermocouple Types

Type	Pair	Range	Note
J	Fe/Constantan	to 750 °C	cheap
K	Chromel/Alumel	to 1260 °C	**most common**
T	Cu/Constantan	to 350 °C	cryogenic
E	Chromel/Constantan	to 900 °C	highest μV/°C
R/S	Pt-Rh/Pt	to 1600 °C	precision, high T
B	PtRh30/PtRh6	to 1800 °C	highest range
CJC	Reference junction at 0 °C assumed; compensate electronically (RTD/IC at terminals)		

Display One-Liners

- **LED:** emits (electroluminescence), λ = 1240/E_g nm, CA segment lights on LOW, CC on HIGH; multiplex digits > 50 Hz.
- **LCD:** modulates light, twisted-nematic 90°, **AC drive only**, μW power, slow (ms), poor in dark without backlight.
- **OLED:** self-emissive organic pixels — true black, thin, flexible; ageing/burn-in.

CRO One-Liners

- Trigger = same point every sweep → stable trace; sweep = linear sawtooth; retrace blanked.
- **Delay line delays vertical signal ≈ 100–200 ns so the triggering edge itself is displayed; delay = n√(LC).**
- **Probe cable is shielded coax to stop external EMI/noise pickup**; 10:1 probe → 10 MΩ, compensate on CAL square wave.
- Dual-trace = one beam switched (ALT/CHOP); dual-beam = two beams; DSO = S/H + ADC + memory → single-shot & pre-trigger capture; beware aliasing.

End of Chapter 6. Next: Chapter 7 as per book plan.